# Appendix

## A: Extra header examples

The example extra header structures below include additional white space and formatting for readability, which should not be used in an actual record.

### FDSN reserved extra headers

A relatively simple example of reserved FDSN extra headers that contains a timing quality value and an event detection is:

```
{
  "FDSN": {
    "Time": {
      "Quality": 100,
      "Correction": 1.234
    },
    "Event": {
      "Begin": true,
      "End": true,
      "InProgress": true,
      "Detection": [
        {
          "Type": "MURDOCK",
          "SignalAmplitude": 80,
          "SignalPeriod": 0.4,
          "BackgroundEstimate": 18,
          "Wave": "DILATATION",
          "Units": "COUNTS",
          "OnsetTime": "2022-06-05T20:32:39.120000Z",
          "MEDSNR": [ 1, 3, 2, 1, 4, 0 ],
          "MEDLookback": 2,
          "MEDPickAlgorithm": 0,
          "Detector": "Z_SPWWSS"
        }
      ]
    }
  }
}
```

An example of the reserved FDSN headers defined by the FDSN is provided below. In this (pathalogical) illustration, all reserved fields are represented.

```
{
  "FDSN": {
    "Time": {
      "Quality": 100,
      "Correction": 1.234,
      "MaxEstimatedError": 1e-06,
      "LeapSecond": 1,
      "Exception": [
        {
          "Time": "2022-05-06T20:32:41.12Z",
          "VCOCorrection": 50.7812,
          "ReceptionQuality": 80,
          "Count": 23,
          "Type": "Valid Timemark",
          "ClockStatus": "SNR=48,51,51,50,50,48,46,48,48,45,45"
        },
        {
          "Time": "2022-05-06T20:32:42.185Z",
          "VCOCorrection": 44.1313,
          "ReceptionQuality": 55,
          "Count": 19690,
          "Type": "Missing timemarks",
          "ClockStatus": "SNR=50,48,46,48,48,45,45"
        }
      ]
    },
    "Event": {
      "Begin": true,
      "End": true,
      "InProgress": true,
      "Detection": [
        {
          "Type": "GENERIC",
          "SignalAmplitude": 80,
          "SignalPeriod": 0.4,
          "BackgroundEstimate": 18,
          "OnsetTime": "2022-05-06T20:32:39.120000Z",
          "Detector": "Dalek STA/LTA"
        },
        {
          "Type": "MURDOCK",
          "SignalAmplitude": 80,
          "SignalPeriod": 0.4,
          "BackgroundEstimate": 18,
          "Wave": "DILATATION",
          "Units": "COUNTS",
          "OnsetTime": "2022-05-06T20:32:39.185000Z",
          "MEDSNR": [ 1, 3, 2, 1, 4, 0 ],
          "MEDLookback": 2,
          "MEDPickAlgorithm": 0,
          "Detector": "Z_SPWWSS"
        }
      ]
    },
    "Calibration": {
      "Sequence": [
        {
          "Type": "Step",
          "BeginTime": "2022-05-06T20:32:39.120000Z",
          "Steps": 12,
          "StepFirstPulsePositive": true,
          "StepAlternateSign": true,
          "Trigger": "AUTOMATIC",
```


latest

```
          "Continued": false,
          "Amplitude": 1345,
          "InputUnits": "COUNTS",
          "Duration": 603.456,
          "SinePeriod": 5.0,
          "StepBetween": 500.0,
          "InputChannel": "CAL",
          "ReferenceAmplitude": 45.8,
          "Coupling": "RESISTIVE",
          "Rolloff": "Description of rolloff"
        },
        {
          "Type": "Step",
          "EndTime": "2022-05-06T20:32:39.120000Z"
        },
        {
          "Type": "Sine",
          "BeginTime": "2022-05-06T20:32:39.120000Z",
          "EndTime": "2022-05-06T20:32:39.120000Z",
          "Trigger": "MANUAL",
          "Continued": true,
          "Amplitude": 1345,
          "InputUnits": "COUNTS",
          "AmplitudeRange": "PEAKTOPEAK",
          "SinePeriod": 5.0,
          "InputChannel": "CAL",
          "ReferenceAmplitude": 45.8,
          "Coupling": "RESISTIVE",
          "Rolloff": "Description of rolloff"
        },
        {
          "Type": "PseudoRandom",
          "BeginTime": "2022-05-06T20:32:39.120000Z",
          "EndTime": "2022-05-06T20:32:39.120000Z",
          "Trigger": "MANUAL",
          "Amplitude": 0.0001,
          "InputUnits": "M/S",
          "Duration": 300,
          "InputChannel": "CAL",
          "ReferenceAmplitude": 45.8,
          "Coupling": "CAPACITIVE",
          "Rolloff": "Very randomly",
          "Noise": "White"
        },
        {
          "Type": "Generic",
          "BeginTime": "2022-05-06T20:32:39.120000Z",
          "EndTime": "2022-05-06T20:32:39.120000Z",
          "Trigger": "MANUAL",
          "Amplitude": 1345,
          "Duration": 100
        }
      ]
    },
    "Recenter": {
      "Sequence": [
        {
          "Type": "Gimbal",
          "BeginTime": "2022-05-06T20:32:39.120000Z",
          "EndTime": "2022-05-06T20:32:39.120000Z",
          "Trigger": "AUTOMATIC"
        },
        {
          "BeginTime": "2022-05-06T20:32:40.120000Z"
```


latest

```
        }
      ]
    },
    "Flags": {
      "MassPositionOffscale": true,
      "AmplifierSaturation": true,
      "DigitizerClipping": true,
      "Spikes": true,
      "Glitches": true,
      "FilterCharging": true,
      "StationVolumeParityError": true,
      "LongRecordRead": true,
      "ShortRecordRead": true,
      "StartOfTimeSeries": true,
      "EndOfTimeSeries": true,
      "MissingData": true,
      "TelemetrySyncError": true
    },
    "Logger": {
      "Model": "DM24",
      "Serial": "A4567"
    },
    "Sensor": {
      "Model": "T240",
      "Serial": "123123"
    },
    "Clock": {
      "Model": "P273T11N16",
      "Serial": "24A00000"
    },
    "ProvenanceURI": "prov:sp001_wf_f84fb9a",
    "DataQuality": "D",
    "Sequence": 123456
  }
}
```


## Non-FDSN extra headers

An example of reserved FDSN headers combined with other top-level headers, illustrating how custom headers may be added.

```
{
  "FDSN": {
    "Time": {
      "Quality": 90
    }
  },
  "Manufacturer123": {
    "Metadata": {
      "FilamentCurrent": 16.4,
      "HyperCoordinates": "1.1789:965402:73324@3.14159"
    }
  },
  "OperatorXYZ": {
    "DSP": {
      "PeakRMS": 2067,
      "RMSWindow": 10.5
    }
  }
}
```


## B: Reference data

The reference data set is intended as an illustration of properly constructed miniSEED 3 and to be used by software implementors during development and testing.

The set contains multiple examples of data records illustrating different characteristics and data encodings of the from. Each example is available in miniSEED 3, JSON, and human-readable text representations.

The whole data set may be downloaded from documentation repository.

All time series in the reference set that contain series are the same expanding sinusoid signal.

| # | Description | Download |
|---|---|---|
| 1 | Text payload | mseed3 JSON Te: |
| 2 | Event detection headers only, no data payload | mseed3 JSON Te: |
| 3 | Sinusoid series encoded as 16-bit integers | mseed3 JSON Te: |
| 4 | Sinusoid series encoded as 32-bit integers | mseed3 JSON Te: |
| 5 | Sinusoid series encoded as 32-bit IEEE float | mseed3 JSON Te: |
| 6 | Sinusoid series encoded as 64-bit IEEE float | mseed3 JSON Te: |
| 7 | Sinusoid series encoded as Steim-1 compressed integers | mseed3 JSON Te: |
| 8 | Sinusoid series encoded as Steim-2 compressed integers | msee latest |
| 9 | Series with time quality, correction, event detections headers | mseed3 JSON Te: |

| # | Description | Download |
|---|---|---|
| 10 | Series with some FDSN and non-FDSN extra headers | mseed3 JSON Te |
| 11 | Series with all FDSN extra headers (unrealistic) | mseed3 JSON Te |

## C: Mapping from miniSEED 2.4

The following list of miniSEED 2.4 format structures specifies the mapping of all fields to this specification. In this listing the following abbreviations are used:

**EH:** Extra Header (of this specification), with path
**SID:** Source Identifier (of this specification)
**FSDH:** Fixed Section Data Header (in either specification)

### miniSEED 2.4 Fixed Section Data Header (FSDH)

<table>
<tr><th>Field</th><th>Description</th><th>This specification</th></tr>
<tr><td>1</td><td>Sequence number</td><td>EH: FDSN.Sequence</td></tr>
<tr><td>2</td><td>Data header/quality indicator</td><td>EH: FDSN.DataQuality<br>A data center may choose to translate miniSEED 2.4 quality values to publication versions with the following mapping:
<table>
<tr><th>2.4 quality</th><th>Pub. version</th></tr>
<tr><td>R</td><td>1</td></tr>
<tr><td>D</td><td>2</td></tr>
<tr><td>Q</td><td>3</td></tr>
<tr><td>M</td><td>4</td></tr>
</table></td></tr>
<tr><td>3</td><td>Reserved byte</td><td>[no mapping]</td></tr>
<tr><td>4</td><td>Station identifier code</td><td>Incorporated into SID, FSDH field 13</td></tr>
<tr><td>5</td><td>Location identifier</td><td>Incorporated into SID, FSDH field 13</td></tr>
<tr><td>6</td><td>Channel identifier</td><td>Incorporated into SID, FSDH field 13</td></tr>
<tr><td>7</td><td>Network code</td><td>Incorporated into SID, FSDH field 13</td></tr>
<tr><td>8</td><td>Record start time</td><td>FSDH field 4</td></tr>
<tr><td>9</td><td>Number of samples</td><td>FSDH field 7</td></tr>
<tr><td>10</td><td>Sample rate factor</td><td>Incorporated into FSDH field 6</td></tr>
</table>

| Field | Description | This specification |
|---|---|---|
| 11 | Sample rate multiplier | Incorporated into FSDH field 6 |
| 12 | **Activity flags:**<br>0 = calibration signals<br>1 = time correction applied<br>2 = begining of event<br>3 = end of event<br>4 = + leap second included<br>5 = - leap second included<br>6 = event in progress | **FSDH and Extra headers:**<br>FSDH field 3, bit 0<br>[no mapping]<br>*FDSN.Event.Begin*<br>*FDSN.Event.End*<br>*FDSN.Time.LeapSecond*<br>*FDSN.Time.LeapSecond*<br>*FDSN.Event.InProgress* |
| 13 | **I/O flags, bits:**<br>0 = Sta. volume parity error<br>1 = Long record read<br>2 = Short record read<br>3 = Start of time series<br>4 = End of time series<br>5 = Clock locked | **FSDH and Extra headers:**<br>*FDSN.Flags.StationVolumeParityError*<br>*FDSN.Flags.LongRecordRead*<br>*FDSN.Flags.ShortRecordRead*<br>*FDSN.Flags.StartOfTimeSeries*<br>*FDSN.Flags.EndOfTimeSeries*<br>FSDH field 3, bit 2 |
| 14 | **Data quality flags, bits:**<br>0 = Amp. saturation detected<br>1 = Dig. clipping detected<br>2 = Spikes detected<br>3 = Glitches detected<br>4 = Missing/padded data<br>5 = Telemetry synch. error<br>6 = Digital filter charging<br>7 = Time tag questionable | **FSDH and Extra headers:**<br>*FDSN.Flags.AmplifierSaturation*<br>*FDSN.Flags.DigitizerClipping*<br>*FDSN.Flags.Spikes*<br>*FDSN.Flags.Glitches*<br>*FDSN.Flags.MissingData*<br>*FDSN.Flags.TelemetrySyncError*<br>*FDSN.Flags.FilterCharging*<br>FSDH field 3, bit 1 |
| 15 | Number of blockettes that follow | [no mapping] |
| 16 | Time correction | EH: *FDSN.Time.Correction* |
| 17 | Beginning of data | [no mapping] |
| 18 | First blockette | [no mapping] |

## Blockette 100 (Sample Rate)

| Field | Description | This specification |
|---|---|---|
| 3 | Actual sample rate | FSDH field 6 |

| Field | Description | This specification |
|---|---|---|
| 4 | Flags (undefined) | [no mapping] |
| 4 | Reserved byte | [no mapping] |

## Blockette 200 (Generic Event Detection)

| Field | Description | This specification |
|---|---|---|
| 3 | Signal amplitude | EH: *FDSN.Event.Detection.SignalAmplitude* |
| 4 | Signal period | EH: *FDSN.Event.Detection.SignalPeriod* |
| 5 | Background estimate | EH: *FDSN.Event.Detection.BackgroundEstimate* |
| 6 | **Event detection flag bits:**<br>0 = dil./comp. wave<br>1 = units | **Extra headers:**<br>*FDSN.Event.Detection.Wave*<br>*FDSN.Event.Detection.Units* |
| 7 | Reserved byte | [no mapping] |
| 8 | Signal onset time | EH: *FDSN.Event.Detection.OnsetTime* |
| 9 | Detector name | EH: *FDSN.Event.Detection.Detector* |

## Blockette 201 (Murdock Event Detection)

| Field | Description | This specification |
|---|---|---|
| 3 | Signal amplitude | EH: *FDSN.Event.Detection.SignalAmplitude* |
| 4 | Signal period | EH: *FDSN.Event.Detection.SignalPeriod* |
| 5 | Background estimate | EH: *FDSN.Event.Detection.BackgroundEstimate* |
| 6 | **Event detection flag bits:**<br>0 = dil./comp. wave | **Extra headers:**<br>*FDSN.Event.Detection.Wave* |
| 7 | Reserved byte | [no mapping] |
| 8 | Signal onset time | EH: *FDSN.Event.Detection.OnsetTime* |
| 9 | Signal-to-noise values | EH: *FDSN.Event.Detection.MEDSNR* (array) |
| 10 | Lookback value | EH: *FDSN.Event.Detection.MEDLookback* |
| 11 | Pick algorithm | EH: *FDSN.Event.Detection.MEDPickAlgorithm* |
| 12 | Detector name | EH: *FDSN.Event.Detection.Detector* |



## Blockette 300 (Step Calibration)

| Field | Description | This specification |
|---|---|---|
| 3 | Start calib | EH: *FDSN.Calibration.Sequence.Begintime* |
| 4 | Num of calibs | EH: *FDSN.Calibration.Sequence.Steps* |
| 5 | **Calibration flags:**<br>0 = first pulse<br>1 = cal alt sign<br>2 = cal auto<br>3 = cal cont | **Extra headers:**<br>*FDSN.Calibration.Sequence.StepFirstPulsePositive*<br>*FDSN.Calibration.Sequence.StepAlternateSign*<br>*FDSN.Calibration.Sequence.Trigger*<br>*FDSN.Calibration.Sequence.Continued* |
| 6 | Duration of step | EH: *FDSN.Calibration.Sequence.Duration* |
| 7 | Time between steps | EH: *FDSN.Calibration.Sequence.StepBetween* |
| 8 | Amp. of cal. signal | EH: *FDSN.Calibration.Sequence.Amplitude* |
| 9 | Channel cal. signal | EH: *FDSN.Calibration.Sequence.InputChannel* |
| 10 | Reserved byte | [no mapping] |
| 11 | Reference amplitude | EH: *FDSN.Calibration.Sequence.ReferenceAmplitude* |
| 12 | Coupling of signal | EH: *FDSN.Calibration.Sequence.Coupling* |
| 13 | Rolloff of filters | EH: *FDSN.Calibration.Sequence.Rolloff* |

## Blockette 310 (Sine Calibration)

| Field | Description | This specification |
|---|---|---|
| 3 | Start calib | EH: *FDSN.Calibration.Sequence.Begintime* |
| 4 | Reserved byte | [no mapping] |
| 5 | **Calibration flags:**<br>2 = cal auto<br>3 = cal cont<br>4 = PtoP amp<br>5 = ZtoP amp<br>6 = RMS amp | **Extra headers:**<br>*FDSN.Calibration.Sequence.Trigger*<br>*FDSN.Calibration.Sequence.Continued*<br>*FDSN.Calibration.Sequence.AmplitudeRange*<br>*FDSN.Calibration.Sequence.AmplitudeRange*<br>*FDSN.Calibration.Sequence.AmplitudeRange* |
| 6 | Duration of cal | EH: *FDSN.Calibration.Sequence.Duration* |
| 7 | Period of signal | EH: *FDSN.Calibration.Sequence.Period* |
| 8 | Amp. of cal. signal | EH: *FDSN.Calibration.Sequence.Amplitude* |
| 9 | Channel cal. signal | EH: *FDSN.Calibration.Sequence.InputChanr* |
| 10 | Reserved byte | [no mapping] |

| Field | Description | This specification |
|---|---|---|
| 11 | Reference amplitude | EH: *FDSN.Calibration.Sequence.ReferenceAmplitude* |
| 12 | Coupling of signal | EH: *FDSN.Calibration.Sequence.Coupling* |
| 13 | Rolloff of filters | EH: *FDSN.Calibration.Sequence.Rolloff* |

## Blockette 320 (Pseudo-random Calibration)

| Field | Description | This specification |
|---|---|---|
| 3 | Start calib | EH: *FDSN.Calibration.Sequence.Begintime* |
| 10 | Reserved byte | [no mapping] |
| 5 | **Calibration flags:**<br>2 = cal auto<br>3 = cal cont<br>4 = Random amps | **Extra headers:**<br>*FDSN.Calibration.Sequence.Trigger*<br>*FDSN.Calibration.Sequence.Continued*<br>*FDSN.Calibration.Sequence.AmplitudeRange* |
| 6 | Duration of cal | EH: *FDSN.Calibration.Sequence.Duration* |
| 7 | PtoP amp. of steps | EH: *FDSN.Calibration.Sequence.Amplitude* |
| 8 | Channel cal. signal | EH: *FDSN.Calibration.Sequence.InputChannel* |
| 9 | Reserved byte | [no mapping] |
| 10 | Reference amplitude | EH: *FDSN.Calibration.Sequence.ReferenceAmplitude* |
| 11 | Coupling of signal | EH: *FDSN.Calibration.Sequence.Coupling* |
| 12 | Rolloff of filters | EH: *FDSN.Calibration.Sequence.Rolloff* |
| 13 | Noise type | EH: *FDSN.Calibration.Sequence.Noise* |

## Blockette 390 (Generic Calibration)

| Field | Description | This specification |
|---|---|---|
| 3 | Start calib | EH: *FDSN.Calibration.Sequence.Begintime* |
| 10 | Reserved byte | [no mapping] |
| 5 | **Calibration flags:**<br>2 = cal auto<br>3 = cal cont | **Extra headers:**<br>*FDSN.Calibration.Sequence.Trigger*<br>*FDSN.Calibration.Sequence.Continued* |
| 6 | Duration of cal | EH: *FDSN.Calibration.Sequence.Duration* |
| 7 | Amplitude of signal | EH: *FDSN.Calibration.Sequence.Amplitude* |

| Field | Description | This specification |
|---|---|---|
| 8 | Channel cal. signal | EH: *FDSN.Calibration.Sequence.InputChannel* |
| 9 | Reserved byte | [no mapping] |

## Blockette 395 (Calibration Abort)

| Field | Description | This specification |
|---|---|---|
| 3 | End calib | EH: *FDSN.Calibration.Sequence.Endtime* |
| 10 | Reserved byte | [no mapping] |

## Blockette 400 (Beam), Blockette 405 (Beam Delay)

No mapping for these blockettes

## Blockette 500 (Timing)

| Field | Description | This specification |
|---|---|---|
| 3 | VCO correction | EH: *FDSN.Time.Exception.VCOCorrection* |
| 4 | Time of exception | EH: *FDSN.Time.Exception.Time* |
| 5 | Microsecond offset | [included in record start time] |
| 6 | Reception quality | EH: *FDSN.Time.Exception.ReceptionQuality* |
| 7 | Exception count | EH: *FDSN.Time.Exception.Count* |
| 8 | Exception type | EH: *FDSN.Time.Exception.Type* |
| 9 | Clock model | EH: *FDSN.Clock.Model* |
| 10 | Clock status | EH: *FDSN.Time.Exception.ClockStatus* |

## Blockette 1000 (Data Only SEED)

| Field | Description | This specification |
|---|---|---|
| 3 | Encoding format | FSDH field 5 |
| 4 | Word order | [no mapping, no longer needed] |
| 5 | Data record length | [no mapping, no longer needed] |
| 6 | Reserved byte | [no mapping] |

## Blockette 1001 (Data Extension)

| Field | Description | This specification |
| --- | --- | --- |
| 3 | Timing quality | EH: *FDSN.Time.Quality* |
| 4 | Microsecond offset | Incorporated into FSDH field 4 |
| 5 | Reserved byte | [no mapping] |
| 6 | Frame count | [no mapping] |

## Blockette 2000 (Variable Length Opaque)

No mapping for this blockette. The opaque data encoding may be used to specify an opaque payload for nearly equivalent functionality.

## Unsupported miniSEED 2.4 content

The following defined information in miniSEED 2.4 cannot be represented in this specification:

- Clock model specification per timing exception. Current specification only allows a single clock model specification per record.
- Blockettes 400 (Beam) & 405 (Beam Delay)
- Blockette 2000 (Opaque Data)